Basic Arithmetic
BOS

NAME:

Q 1 **2000**

Solve $|x-5|=3$.

$$|x-5| = 3$$
$$x-5 = 3 \quad \text{or} \quad -(x-5) = 3$$
$$\therefore \quad x = 8. \qquad x-5 = -3$$
$$\therefore \quad x = 2.$$

Solution: $x = 8$ or $x = 2$.

Q 2 **2001**

Evaluate, correct to three significant figures,

$$\sqrt{\frac{3^2+12^2}{231-12^2}}$$

$$\sqrt{\frac{3^2+12^2}{231-12^2}} = \sqrt{\frac{153}{87}}$$
$$= 1{\cdot}326\,12$$
$$= 1{\cdot}33 \text{ (correct to 3 sig. figs).}$$

Q 3 **2001**

Solve $|x+3|<2$.

Graph your solution on a number line.

(b) $|x+3| < 2$
$$-2 < x+3 < 2$$
$$-5 < x < -1.$$

OR $x+3 < 2$ or $-x-3 < 2$
$$x < -1 \quad \text{or} \quad -x < 5$$
$$x > -5.$$
$$\therefore \quad -5 < x < -1.$$

Q 4 **2001**

Solve $x^2-2x-8=0$.

(c)
$$x^2-2x-8 = 0$$
$$(x-4)(x+2) = 0$$
$$x = 4 \quad \text{or} \quad x = -2.$$

OR
$$x = \frac{2\pm\sqrt{4+32}}{2}$$
$$= \frac{2\pm 6}{2}$$
$$= 4 \text{ or } -2.$$

Q 5 **2001**

Simplify $\frac{x}{x^2-4}+\frac{2}{x-2}$

$$\frac{x}{x^2-4}+\frac{2}{x-2} = \frac{x}{(x-2)(x+2)}+\frac{2}{x-2}$$
$$= \frac{x+2(x+2)}{(x-2)(x+2)}$$
$$= \frac{3x+4}{(x-2)(x+2)}$$
$$= \frac{3x+4}{x^2-4}.$$

Q 6 **2001**

The cost of a video recorder is \$979. This includes a 10% tax on the original price. Calculate the original price of the video recorder.

Let \$$x$ be the original cost.
$$x + 10\% \text{ of } x = 979$$
$$x + 0{\cdot}1x = 979$$
$$1{\cdot}1x = 979$$
$$x = 890.$$

$\therefore$ The original cost is \$890.

Q 7 **2002**

Evaluate, correct to three significant figures,

$$\frac{5.8^2 - 3.1^3}{3 \times 3.1 \times 5.8}.$$

$$\begin{aligned} \frac{5.8^2 - 3.1^3}{3 \times 3.1 \times 5.8} &= \frac{33.64 - 29.791}{53.94} \\ &= \frac{3.849}{53.94} \\ &= 0.07135\ldots \quad &\text{(by calc.)} \\ &= 0.0714 \quad &\text{to 3 sig. figures} \end{aligned}$$

Q 8 **2002**

Solve $x^2 = 5x$.

$$\begin{aligned} x^2 &= 5x \\ \therefore x^2 - 5x &= 0 \\ \therefore x(x-5) &= 0 \\ \therefore x &= 0 \text{ or } 5. \end{aligned}$$

Q 9 **2002**

Solve $3x - \dfrac{2x-5}{2} = 6$

$$\begin{aligned} 3x - \frac{2x-5}{2} &= 6 \\ \therefore 6x - (2x-5) &= 12 \\ 6x - 2x + 5 &= 12 \\ 4x + 5 &= 12 \\ 4x &= 7 \\ x &= \frac{7}{4}. \end{aligned}$$

Q10 **2002**

Solve the pair of simultaneous equations

$$x - 2y = 8$$
$$2x + y = 1$$

$$\begin{array}{lrl} & x - 2y = 8 & \ldots (1) \\ & 2x + y = 1 & \ldots (2) \\ (2) \times 2 & 4x + 2y = 2 & \ldots (3) \\ (1) + (3) & 5x = 10 & \\ & \therefore x = 2 & \end{array}$$

Substituting $x = 2$ into (2):

$$\begin{aligned} 2(2) + y &= 1 \\ 4 + y &= 1 \\ y &= -3. \end{aligned}$$

Q11 **2003**

Josh invests \$1000 in a term deposit that earns 3.5% per annum compounded annually.

What is the value of the investment at the end of 20 years?

$$\begin{aligned} A &= P(1+r)^n \\ &= 1000\left(1 + \frac{3.5}{100}\right)^{20} \\ &= 1000(1 + 0.035)^{20} \\ &= 1000(1.035)^{20} \\ &= \$1989.79 \quad \text{to nearest cent.} \end{aligned}$$

Q12 **2002**

Solve $|x-1| \geq 3$ and graph your solution on the number line.

$$|x-1| \geq 3$$
$$\therefore x-1 \geq 3 \quad \text{or} \quad -(x-1) \geq 3$$
$$\therefore x \geq 4 \quad \text{or} \quad -x+1 \geq 3$$
$$\therefore x \geq 4 \quad \text{or} \quad x \leq -2$$

Q13 **2003**

Evaluate $e^{-3.5}$ correct to 3 significant figures.

$e^{-3.5} = 0.03019\ldots$ (by calc.)
$= 0.0302$ to 3 sig. figures.

Q14 **2003**

Iain and Louise paid \$315.00 for a meal at a restaurant. This included a $12\frac{1}{2}\%$ tip. What was the cost of the meal without the tip?

original cost + $12\frac{1}{2}\%$ of original cost = final cost
Let x represent the original price.

$$\therefore x + \frac{12\frac{1}{2}}{100}x = \$315$$
$$\therefore x + 0.125x = \$315$$
$$\therefore 1.125x = \$315$$
$$\therefore x = \frac{\$315}{1.125}$$
$$= \$280. \quad \text{(by calc.)}$$

$\therefore$ The cost of the meal without tip was \$280.

Q15 **2003**

Solve $|x-3| = 7$.

$$|x-3| = 7$$
$$\therefore x-3 = \pm 7$$
$$\therefore x-3 = 7 \quad \text{or} \quad x-3 = -7$$
$$\therefore x = 10 \quad \text{or} \quad x = -4.$$

Q16 **2004**

The radius of Mars is approximately 3 397 000 m. Write this number in scientific notation, correct to two significant figures.

$$3\,397\,000 = 3{\cdot}397 \times 10^6$$
$$= 3{\cdot}4 \times 10^6 \text{ (2 significant figures).}$$

Q17 **2004**

Solve $\frac{x-5}{3} - \frac{x+1}{4} = 5$.

$$\frac{x-5}{3} - \frac{x+1}{4} = 5.$$
$$12 \times \frac{(x-5)}{3} - 12 \times \frac{(x+1)}{4} = 12 \times 5$$
$$4(x-5) - 3(x+1) = 60$$
$$4x - 20 - 3x - 3 = 60$$
$$x - 23 = 60$$
$$x = 83.$$

Q18 **2004**

Find integers a and b such that $(3-\sqrt{2})^2 = a - b\sqrt{2}$.

$$\begin{aligned}(3-\sqrt{2})^2 &= 9 - 2(3)\sqrt{2} + (\sqrt{2})^2\\ &= 9 - 6\sqrt{2} + 2\\ &= 11 - 6\sqrt{2}.\end{aligned}$$

Need to find values a, b such that

$(3-\sqrt{2})^2 = a - b\sqrt{2}$.

$\therefore\ a = 11,\ b = 6$.

Q19 **2004**

Find the values of x for which $|x+1| \le 5$.

$|x+1| \le 5$.

METHOD 1 $\quad x+1 \le 5$ or $-(x+1) \le 5$

$x \le 4 \qquad x+1 \ge -5$

$x \ge -6$.

$\therefore -6 \le x \le 4$.

METHOD 2

$|x+1| \le 5. \qquad -5 \le x+1 \le 5$

$\therefore \qquad -6 \le x \le 4$.

Q20 **2005**

Evaluate $\sqrt{\dfrac{275.4}{5.2 \times 3.9}}$ correct to two significant figures.

$$\begin{aligned}\sqrt{\frac{275\cdot4}{5\cdot2\times3\cdot9}} &= 3\cdot685\\ &= 3\cdot7 \text{ (2 sig. figs).}\end{aligned}$$

Q21 **2005**

Factorise $x^3 - 27$.

$$\begin{aligned}x^3 - 27 &= x^3 - 3^3\\ &= (x-3)(x^2+3x+9).\end{aligned}$$

Q22 **2005**

Express $\dfrac{(2x-3)}{2} - \dfrac{(x-1)}{5}$ as a single fraction in its simplest form.

$$\begin{aligned}\frac{(2x-3)}{2} - \frac{(x-1)}{5} &= \frac{5(2x-3) - 2(x-1)}{10}\\ &= \frac{10x - 15 - 2x + 2}{10}\\ &= \frac{8x-13}{10}.\end{aligned}$$

Q23 **2005**

Find the values of x for which $|x-3| \le 1$.

$|x-3| \le 1$.

METHOD 1 $\qquad |x-3| \le 1$

$-1 \le x-3 \le 1$

$\therefore \qquad 2 \le x \le 4$.

METHOD 2 $\qquad |x-3| \le 1$

$x-3 \le 1$ or $-(x-3) \le 1$

$x \le 4 \qquad x-3 \ge -1$

$x \ge -2$.

$\therefore\ 2 \le x \le 4$.

Q24 **2006**

Evaluate $e^{-0.5}$ correct to three decimal places.

$\underline{0.6065\ldots=0.607}(3dp)$

Q25 **2006**

Factorise $2x^2+5x-3$.

$2x^2+5x-3=(2x-1)(x+3)$

Q26 **2006**

Solve $3-5x \le 2$.

$3-5x \le 2 \Rightarrow -5x \le -1 \Rightarrow x \ge \frac{1}{5}$

Q27 **2007**

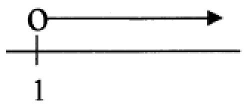
Evaluate $\sqrt{\pi^2+5}$ correct to two decimal places.

$=3.856=3.86$

Q28 **2007**

Solve $2x-5>-3$ and graph the solution on a number line.

$$2x-5>-3$$
$$2x>2$$
$$x>1$$



Q29 **2007**

Rationalise the denominator of $\frac{1}{\sqrt{3}-1}$.

$$\frac{1}{\sqrt{3}-1}=\frac{1}{\sqrt{3}-1}\times\frac{\sqrt{3}+1}{\sqrt{3}+1}$$
$$=\frac{\sqrt{3}+1}{3-1}=\frac{\sqrt{3}+1}{2}$$

Q30 **2007**

Factorise $2x^2+5x-12$.

$$2x^2+5x-12$$
$$=(2x-3)(x+4)$$

Q31 **2008**

Factorise $3x^2 + x - 2$.

$$3x^2 + x - 2 = (3x-2)(x+1)$$

Q32 **2008**

Simplify $\frac{2}{n} - \frac{1}{n+1}$.

$$\frac{2}{n} - \frac{1}{n+1} = \frac{2(n+1)-n}{n(n+1)} = \frac{n+2}{n(n+1)}$$

Q33 **2008**

Solve $|4x-3| = 7$.

Q1d $|4x-3| = 7 \Rightarrow 4x-3 = 7$ or $-(4x-3) = 7$,

i.e. $x = \frac{5}{2}$ or -1

Q34 **2008**

Expand and simplify $(\sqrt{3}-1)(2\sqrt{3}+5)$.

$$(\sqrt{3}-1)(2\sqrt{3}+5) = 6 + 5\sqrt{3} - 2\sqrt{3} - 5 = 3\sqrt{3} + 1$$

Q35 **2009**

Solve $\frac{5x-4}{x} = 2$.

$\frac{5x-4}{x} = 2$, $5x - 4 = 2x$, $3x = 4$, $x = \frac{4}{3}$.

Q36 **2009**

Solve $|x+1| = 5$.

$|x+1| = 5$, $x+1 = \pm 5$, $x = -6$ or 4.

Q37 **2010**

Solve $x^2 = 4x$.

$x^2 = 4x$, $x^2 - 4x = 0$, $x(x-4) = 0$, $x = 0$ or 4

Q38 **2010**

Find integers a and b such that $\frac{1}{\sqrt{5}-2} = a + b\sqrt{5}$.

$$a + b\sqrt{5} = \frac{1}{\sqrt{5}-2} \times \frac{\sqrt{5}+2}{\sqrt{5}+2} = 2 + \sqrt{5} \text{ , .: } a = 2 \text{ , } b = 1$$

Q39 **2010**

Solve $|2x+3|=9$.

$|2x+3|=9$, $2x+3=-9$ or $2x+3=9$, $\therefore$ $x=-6$ or 3

Q40 **2010**

Solve the inequality $x^2-x-12<0$.

$x^2-x-12<0$, $(x+3)(x-4)<0$, $-3<x<4$

Q41 **2011**

Evaluate $\sqrt[3]{\frac{651}{4\pi}}$ correct to four significant figures.

$\sqrt[3]{\frac{651}{(4\pi)}} \approx 3.728$ by calculator.

Q42 **2011**

Simplify $\frac{n^2-25}{n-5}$.

$$\frac{n^2-25}{n-5}=\frac{(n-5)(n+5)}{n-5}=n+5$$

Q43 **2011**

Solve $2-3x \le 8$.

$2-3x\le 8$, $-3x\le 6$, $x \ge \frac{6}{-3}$, $x\ge -2$

Q44 **2011**

Rationalise the denominator of $\frac{4}{\sqrt{5}-\sqrt{3}}$

Give your answer in the simplest form.

$$\frac{4}{\sqrt{5}-\sqrt{3}}=\frac{4}{(\sqrt{5}-\sqrt{3})}\times\frac{(\sqrt{5}+\sqrt{3})}{(\sqrt{5}+\sqrt{3})}=\frac{4(\sqrt{5}+\sqrt{3})}{2}=2(\sqrt{5}+\sqrt{3})$$

Q45 **2012**

What is 4.097 84 correct to three significant figures?

(A) 4.09

$\boxed{\text{(B) } 4.10}$

(C) 4.097

(D) 4.098

Q46 **2012**

Which of the following is equal to $\frac{1}{2\sqrt{5}-\sqrt{3}}$?

(A) $\frac{2\sqrt{5}-\sqrt{3}}{7}$

(B) $\frac{2\sqrt{5}+\sqrt{3}}{7}$

(C) $\frac{2\sqrt{5}-\sqrt{3}}{17}$

$\boxed{\text{(D) } \frac{2\sqrt{5}+\sqrt{3}}{17}}$

Q47 **2012**

Factorise $2x^2 - 7x + 3$.

$$2x^2 - 7x + 3 = (2x-1)(x-3)$$

Q48 **2012**

Solve $|3x - 1| < 2$.

Q11b $|3x-1| < 2$

$(3x-1)^2 < 4$, $9x^2 - 6x + 1 < 4$

.: $9x^2 - 6x - 3 < 0$, $3x^2 - 2x - 1 < 0$, $(3x+1)(x-1) < 0$

$\left(x+\frac{1}{3}\right)(x-1) < 0$, .: $-\frac{1}{3} < x < 1$

Q49 **2013**

What are the solutions of $2x^2 - 5x - 1 = 0$?

(A) $x = \dfrac{-5 \pm \sqrt{17}}{4}$

(B) $x = \dfrac{5 \pm \sqrt{17}}{4}$

(C) $x = \dfrac{-5 \pm \sqrt{33}}{4}$

(D) $x = \dfrac{5 \pm \sqrt{33}}{4}$

Q50 **2014**

Which expression is a factorisation of $8x^3 + 27$?

(A) $(2x-3)(4x^2 + 12x - 9)$

(B) $(2x+3)(4x^2 - 12x + 9)$

(C) $(2x-3)(4x^2 + 6x - 9)$

(D) $(2x+3)(4x^2 - 6x + 9)$

Q51 **2014**

Rationalise the denominator of $\dfrac{1}{\sqrt{5}-2}$.

$$\frac{1}{\sqrt{5}-2} = \frac{\sqrt{5}+2}{(\sqrt{5}-2)(\sqrt{5}+2)}$$
$$= \sqrt{5}+2$$

Q52 **2014**

Factorise $3x^2 + x - 2$.

$$3x^2 + x - 2 = (3x-2)(x+1)$$

Q53 **2015**

What is 0.005 233 59 written in scientific notation, correct to 4 significant figures?

(A) 5.2336×10^{-2}

(B) 5.234×10^{-2}

(C) 5.2336×10^{-3}

(D) 5.234×10^{-3}

Q54 **2015**

Simplify $4x - (8 - 6x)$.

$$4x - (8-6x) = 4x - 8 + 6x = 10x - 8$$

Q55
2015

Factorise fully $3x^2 - 27$.

$3x^2 - 27 = 3\left(x^2 - 9\right) = 3(x+3)(x-3)$

Q56
2015

Express $\dfrac{8}{2+\sqrt{7}}$ with a rational denominator.

$$\begin{aligned}\frac{8}{2+\sqrt{7}} &= \frac{8}{2+\sqrt{7}} \times \frac{2-\sqrt{7}}{2-\sqrt{7}} \\ &= \frac{16-8\sqrt{7}}{-3} \\ &= \frac{8\sqrt{7}-16}{3}\end{aligned}$$